RoboSim Navigation Trainer

RoboSim Navigation Trainer is an educational robot navigation simulator that lets students explore robot control, sensor processing, and machine-learning training. It provides a 2-D pygame visualisation of a robot equipped with whisker-style range sensors, tools to record interaction logs, and GUIs for training imitation-learning (MLP) and reinforcement-learning (PPO) models. The suite also includes a benchmark generator and evaluator to compare model performance on standardized maps. Designed for students and researchers, the project demonstrates the full data-flow pipeline from simulation to model deployment.

Getting Started

1. Install the required packages (`pip install -r requirements.txt`).
2. Run `python UTILITY_LAUNCHER.py` to open the Utility Launcher.
3. Click **Open Simulator** to start the navigation simulator, then use the toolbar buttons to control the robot and record logs.
4. Use **Open MLP Trainer** or **Open Benchmark Utility** to train and evaluate models.

Installation

RoboSim Navigation Trainer requires Python 3.8+ and the following packages:

```
pip install -r requirements.txt
```

The `requirements.txt` typically includes: - `pygame` — for the simulator rendering - `numpy` — for numerical computations - `torch` — for PPO neural networks (if using train_ppo.py) - `matplotlib` — for training plots - `gym` — for the headless environment interface

If you plan to use the PPO trainer with GPU acceleration, ensure you have a CUDA-compatible GPU and install the appropriate `torch` version with CUDA support.

Note: The project uses `pygame-ce` (Community Edition), which is a fork of pygame with continued maintenance.

Project Structure

The project is organized as follows:

```
RoboSim_Navigation_Trainer/
├── UTILITY_LAUNCHER.py    # Desktop hub for launching all tools
├── simulator.py            # Main pygame-based navigation simulator
├── sim_core.py             # Core robot physics and obstacle generation
├── train_mlp.py            # MLP imitation learning trainer (Tkinter GUI)
├── train_ppo.py            # PPO reinforcement learning trainer (Tkinter GUI)
├── benchmark_gui.py        # Benchmark map generator and model evaluator
├── headless_sim.py         # Non-GUI simulation environment for training
├── nav_env.py              # OpenAI Gym-compatible environment wrapper
├── requirements.txt         # Python dependencies
├── logs/                   # Directory for simulator JSONL log files
├── trained_models/         # Output directory for saved model JSON files
├── benchmark/              # Benchmark map files for standardized evaluation
├── dagger_snapshots/       # Snapshot directory for DAgger iterations
├── images/                 # Simulator image assets (if any)
└── Docs3/                  # Auto-generated documentation (this README lives here)
```

Key modules for students: - `simulator.py` + `sim_core.py` — Understand robot kinematics, whisker sensor math, and environment generation. - `train_mlp.py` — Study how imitation learning converts logs into neural network weights. - `train_ppo.py` — Explore policy gradient methods and actor-critic architectures.

Application architecture

The diagram below shows how data flows between the main modules of RoboSim Navigation Trainer.

Core Concepts

Before using the tools, understand these key concepts:

Whisker sensors — The robot is equipped with 11 virtual whiskers (like a cat's whiskers) spaced at angles from -90° to +90°. Each whisker measures the distance to the nearest obstacle (up to 0.5 m). These distances are the primary input to the navigation models.

Control modes — The robot supports three control schemes: - *Heading+Drive*: Outputs are `drive_speed` (forward/back) and `rotation_rate` (turning speed). - *XY Strafe*: Outputs are `vx` and `vy` (velocity in x/y directions). - *Hdg+Strafe*: Combined outputs: `rotation_rate`, `vx`, `vy`.

Imitation Learning (IL) — Also called *learning from demonstration*. You drive the robot manually while it records (whisker readings → your actions). A neural network then learns to mimic your driving behavior.

MLP (Multi-Layer Perceptron) — A feedforward neural network with one or more hidden layers. The `train_mlp.py` tool trains an MLP using your recorded logs.

PPO (Proximal Policy Optimization) — A reinforcement learning algorithm that improves the model by running many parallel simulations, collecting rewards, and updating the policy. PPO can start from (warm-start) an IL model and fine-tune it.

DAgger — Dataset Aggregation, an iterative process where the model drives, the human corrects mistakes, and the model retrains on the combined data. The simulator supports DAgger snapshots.

JSONL logs — Each line is a JSON object containing a history of whisker lengths, target bearing, and the action taken. These files are the training data for IL.

Benchmark maps — Standardized obstacle layouts generated with a fixed random seed. They allow fair comparison between different models.

Launch and navigate the simulator

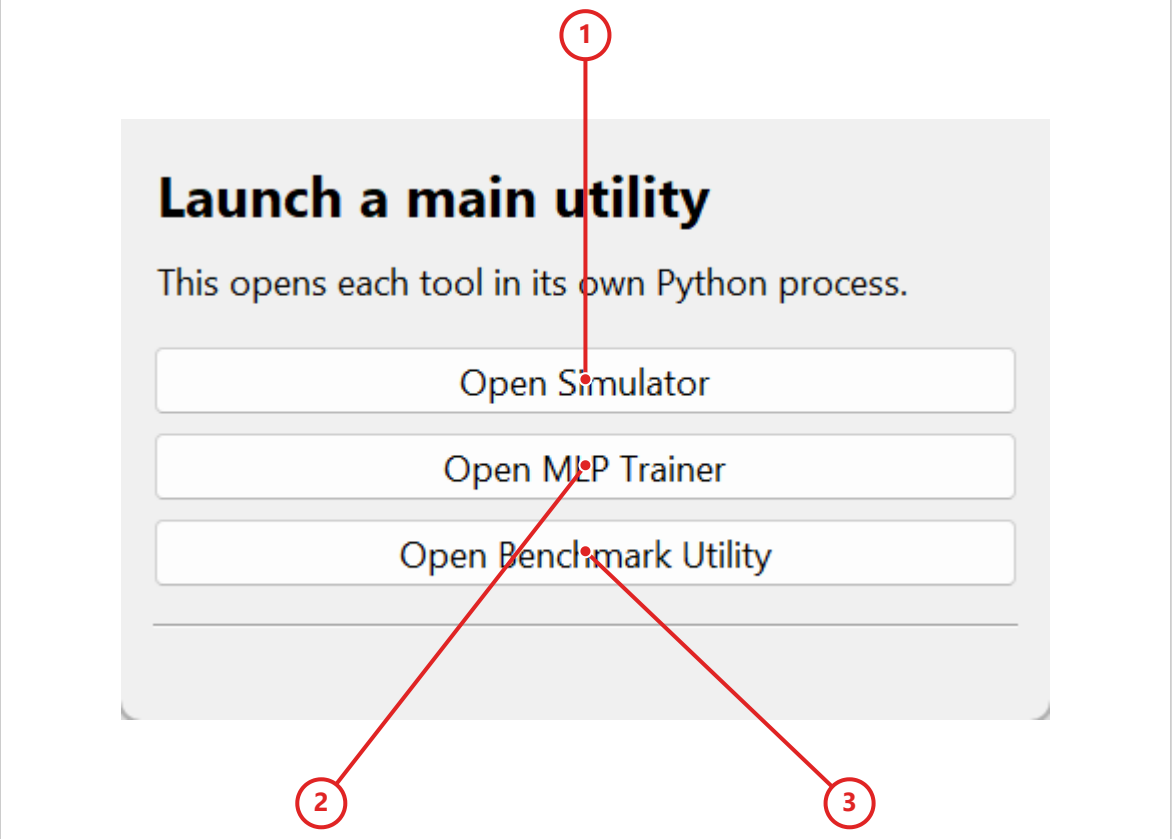
Start the robot navigation simulator, drive the robot using keyboard or gamepad, and generate training logs for imitation learning.

How to use

1. Run `python UTILITY_LAUNCHER.py` and click **Open Simulator** (1) to launch the simulator.
2. In the **Simulator Main Window**, click **Run** (1) to start a new episode with random robot pose and obstacles.
3. Select input mode: **Keyboard** (5) for WASD/arrow keys, **Gamepad** (6) for joystick, or **Model** (7) for autonomous mode.
4. Drive the robot toward the green target while avoiding gray obstacles. Whisker readouts (17) show distances to obstacles.
5. Toggle **Enable Logging** (8) to record sensor/action data to JSONL files for later training.
6. When finished, click **Fail** (2) to end the episode and save logs to the `logs/` directory.

Utility Launcher

Hub window for launching all RoboSim tools in separate processes.



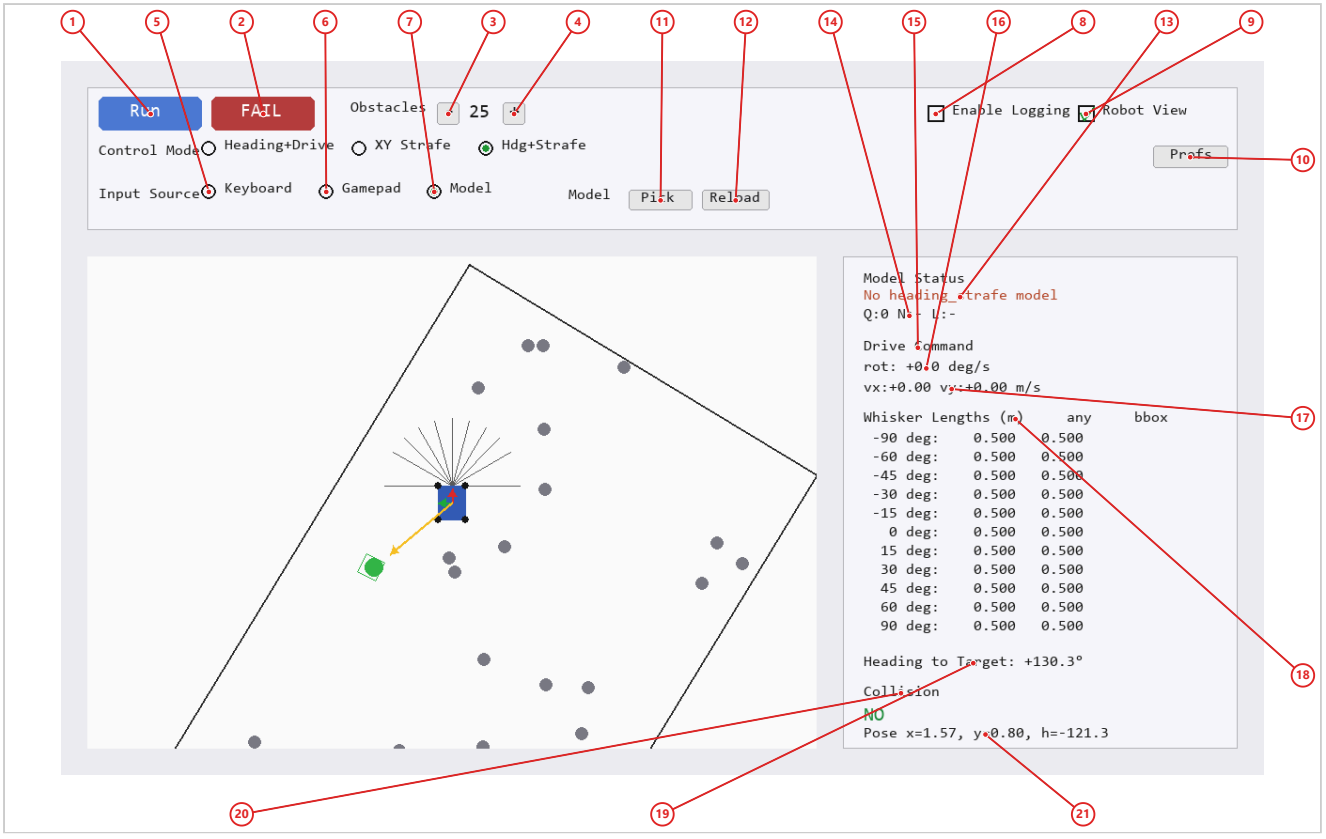
Labeled controls and displays:

1. **Open Simulator** — Launches simulator.py in a new Python process.
2. **Open MLP Trainer** — Launches train_mlp.py in a new Python process.
3. **Open Benchmark Utility** — Launches benchmark_gui.py in a new Python process.

Click **Open Simulator** (1) to start the navigation simulator. Click **Open MLP Trainer** (2) to train imitation learning models from logs. Click **Open Benchmark Utility** (3) to generate benchmark maps and evaluate model performance. The status bar at the bottom shows which tool was last launched.

Simulator Main Window

Pygame-based robot navigation simulation with real-time controls and sensor readouts.



Labeled controls and displays:

1. **Run button** — Starts a new navigation episode with random robot pose and obstacles.
2. **Fail button** — Terminates the current episode and writes logs if logging is enabled.
3. **Obstacle minus** — Decreases the number of obstacles in the simulation by one.
4. **Obstacle plus** — Increases the number of obstacles in the simulation by one.
5. **Keyboard radio** — Switches input to keyboard control (WASD/arrow keys).
6. **Gamepad radio** — Switches input to connected gamepad/joystick.
7. **Model radio** — Switches to autonomous mode using a loaded IL/RL model.
8. **Enable Logging** — Toggles recording of sensor/action data to JSONL log files.

9. **Robot View** — When checked, the simulation view rotates to follow the robot heading.
10. **Preferences** — Opens the preferences dialog for advanced simulation settings.
11. **Model Pick** — Opens file dialog to select a trained model JSON file.
12. **Model Reload** — Reloads the currently selected model from disk.
13. **Model Status** — Shows which control-mode model is loaded or 'No model' if none.
14. **Episode counters** — Displays current episode/step counts (Q=episodes, N=steps, L=logs).
15. **Drive Command** — Section header for the current control-mode outputs.
16. **Rotation rate** — Live readout of robot's current turning speed in deg/s.
17. **Velocity readout** — Live readout of robot's linear velocity components in m/s.
18. **Whisker readouts** — Section header for the 11 whisker distance measurements to obstacles.
19. **Heading to Target** — Current bearing from robot to target in degrees (positive = clockwise).
20. **Collision status** — Displays 'YES' if robot is currently colliding with an obstacle or wall.
21. **Robot Pose** — Live readout of robot position (x,y) in meters and heading in degrees.

Use **Run** (1) to start a new episode and **Fail** (2) to terminate the current episode. Adjust obstacle count with the +/- buttons (3-4). Select input source: **Keyboard** (5), **Gamepad** (6), or **Model** (7). Toggle **Enable Logging** (8) to record training data and **Robot View** (9) to align display with robot heading. Open **Preferences** (10) to adjust advanced settings. When in Model mode, use **Pick** (11) to select a trained model and **Reload** (12) to refresh. The right panel shows live readouts: model status (13), episode/step counters (14), drive commands (15-16), whisker lengths (17), heading to target (18), collision status (19), and robot pose (20).

Train an imitation learning model (MLP)

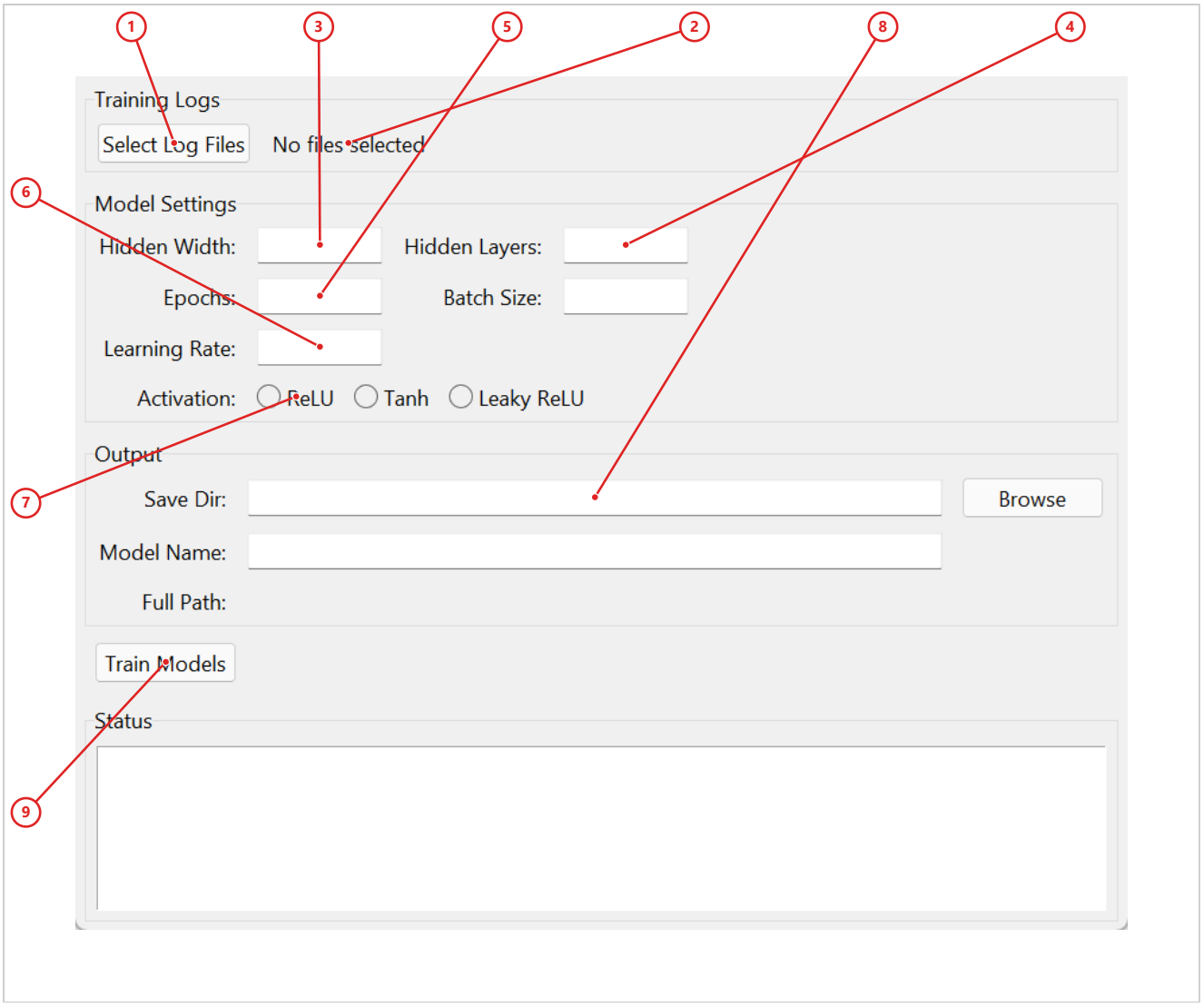
Use the MLP Trainer to teach a neural network to mimic your navigation behavior from recorded logs.

How to use

1. First, generate training logs by running the simulator with logging enabled (see 'Launch and navigate the simulator' workflow).
2. Run `python UTILITY_LAUNCHER.py` and click **Open MLP Trainer** (2) to open the **MLP Trainer**.
3. In the **MLP Trainer**, click **Select Log Files** (1) and choose the JSONL files from the `logs/` directory.
4. Configure network architecture: set **Hidden Width** (3), **Hidden Layers** (4), **Epochs** (5), **Batch Size**, and **Learning Rate** (6).
5. Choose an activation function: **ReLU**, **Tanh**, or **Leaky ReLU** (7).
6. Set the **Save Dir** (8) and optional **Model Name**, then click **Train Models** (9) to start training.
7. The status area shows training progress. When complete, model JSON files are saved to the output directory.

MLP Trainer

GUI for training multi-layer perceptron models from navigation logs using imitation learning.



Labeled controls and displays:

1. **Select Log Files** — Opens file dialog to pick JSONL navigation log files.
2. **File count** — Shows how many log files are currently selected.
3. **Hidden Width** — Number of neurons in each hidden layer of the MLP.
4. **Hidden Layers** — Number of hidden layers in the neural network.
5. **Epochs** — Number of complete passes through the training dataset.
6. **Learning Rate** — Step size for gradient descent during training.
7. **ReLU radio** — Selects ReLU activation for hidden layers.
8. **Save Dir** — Folder where trained model JSON and metrics are saved.
9. **Train Models** — Starts imitation learning training for all modes found in logs.

Click **Select Log Files** (1) to choose one or more JSONL log files. The file count appears in the label (2). Configure the network architecture: **Hidden Width** (3) sets neurons per layer, **Hidden Layers** (4) sets depth. Set **Epochs** (5) and **Learning Rate** (6), then choose an activation function via the radio buttons (7). Specify the **Save Dir** (8) and optional **Model Name**, then click **Train Models** (9) to start training. The status area shows progress and saved file paths.

Fine-tune with PPO reinforcement learning

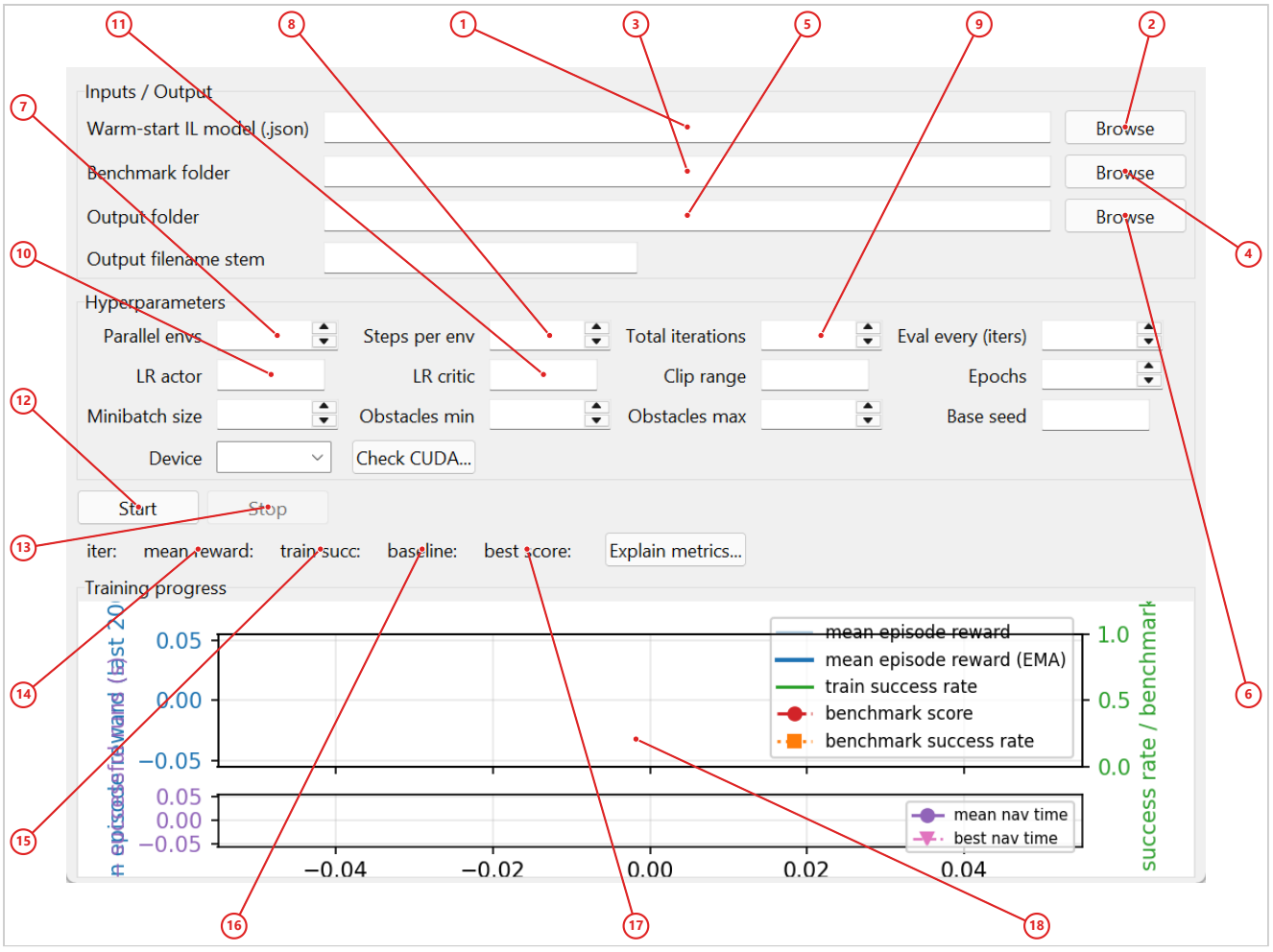
Improve a trained imitation learning model using Proximal Policy Optimization in headless simulation.

How to use

1. Run `python train_ppo.py` to open the **PPO Trainer**.
2. Enter the path to a trained IL model in **Warm-start IL model** (1) or click **Browse** (2) to select it.
3. Set **Benchmark folder** (3) and **Output folder** (4-5) for evaluation and model saving.
4. Configure hyperparameters: **Parallel envs** (7), **Steps per env** (8), **Total iterations** (9), learning rates (10-11), and others.
5. Click **Start** (12) to begin PPO training. The live plot (18) shows reward, success rates, and benchmark scores.
6. Monitor the readouts: **mean reward** (13), **train succ** (14), **baseline** (15), and **best score** (16).
7. Click **Stop** (13) to halt training early. The best model is saved automatically.

PPO Trainer

GUI for fine-tuning navigation models using Proximal Policy Optimization with live plots.



Labeled controls and displays:

1. **Warm-start model** — Path to a pre-trained IL model JSON to initialize PPO.
2. **Browse warm-start** — Opens file dialog to pick IL model JSON file.
3. **Benchmark folder** — Directory containing benchmark map files for evaluation.
4. **Browse benchmark** — Opens directory chooser for benchmark maps.
5. **Output folder** — Directory where PPO model checkpoints will be saved.
6. **Browse output** — Opens directory chooser for output models.
7. **Parallel envs** — Number of parallel simulation environments for training.
8. **Steps per env** — Simulation steps per environment per PPO iteration.
9. **Total iterations** — Number of PPO training iterations to run.
10. **LR actor** — Learning rate for the PPO actor network.
11. **LR critic** — Learning rate for the PPO critic network.
12. **Start button** — Starts the PPO training loop in a background thread.
13. **Stop button** — Stops the currently running training loop.
14. **Mean reward** — Live readout of mean episode reward (last 200 episodes).
15. **Train success** — Live readout of training success rate (0-1).

16. **Baseline score** — Benchmark score of the warm-start IL model.
17. **Best score** — Best PPO benchmark score achieved so far.
18. **Training plot** — Live matplotlib plot of reward, success rates, and benchmark scores.

Enter the **Warm-start IL model** path (1) or browse (2) to load a pre-trained imitation learning model. Set **Benchmark folder** (3) and **Output folder** (4-5). Configure hyperparameters: **Parallel envs** (7), **Steps per env** (8), **Total iterations** (9), learning rates (10-11), **Clip range** (12), **Epochs** (13), **Minibatch size** (14), obstacle counts (15-16), and **Base seed** (17). Choose **Device** (18) and click **Start** (12) to begin PPO training. The live plot (18) shows reward, success rates, and benchmark scores.

Benchmark and evaluate models

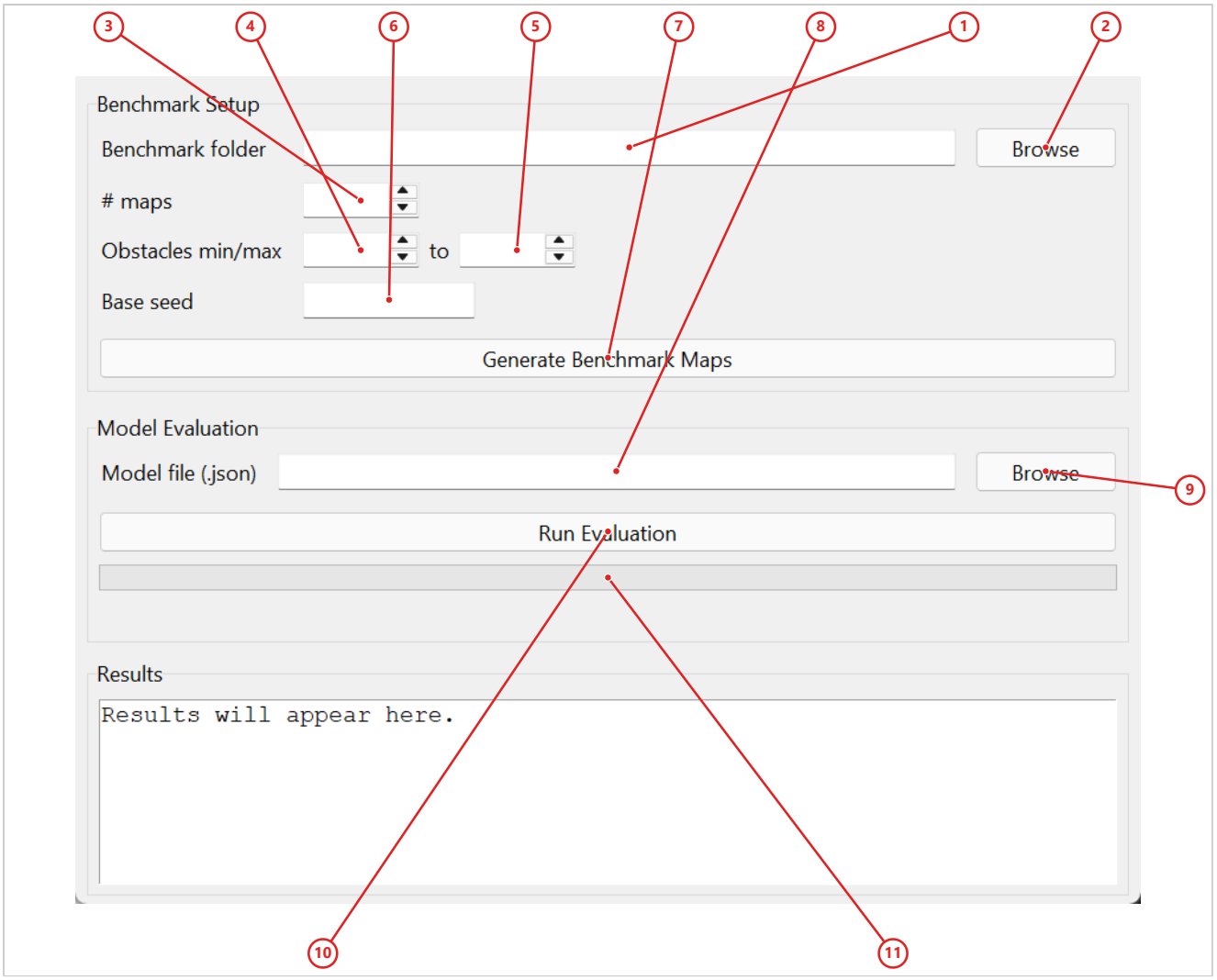
Generate standardized test maps and measure model performance with success rates and navigation times.

How to use

1. Run `python UTILITY_LAUNCHER.py` and click **Open Benchmark Utility** (3) to open the **Benchmark Evaluator**.
2. In the **Benchmark Setup** section, set **Benchmark folder** (1) and number of **# maps** (3) to generate.
3. Configure obstacle range with **Obstacles min/max** (4-5) and **Base seed** (6), then click **Generate Benchmark Maps** (7).
4. For evaluation, enter the **Model file** path (8) or click **Browse** (9) to select a trained model JSON.
5. Click **Run Evaluation** (10). Progress appears in the bar (11) and detailed results show in the text area.
6. Results include success rate, mean navigation time, and benchmark score for the model.

Benchmark Evaluator

GUI for generating benchmark maps and evaluating model performance on standardized scenarios.



Labeled controls and displays:

1. **Benchmark folder** — Directory where benchmark map files are stored.
2. **Browse benchmark** — Opens directory chooser for benchmark maps.
3. **# maps** — Number of benchmark maps to generate.
4. **Obstacles min** — Minimum number of obstacles per benchmark map.
5. **Obstacles max** — Maximum number of obstacles per benchmark map.
6. **Base seed** — Random seed for reproducible map generation.
7. **Generate Maps** — Creates benchmark maps with specified parameters.
8. **Model file** — Path to trained model JSON file for evaluation.
9. **Browse model** — Opens file dialog to select model JSON.
10. **Run Evaluation** — Evaluates model on all benchmark maps and reports metrics.
11. **Progress bar** — Shows evaluation progress across benchmark maps.

Enter the **Benchmark folder** (1) or browse (2) to set where maps are stored. Set **# maps** (3) to generate, plus obstacle min/max (4-5) and **Base seed** (6). Click **Generate Benchmark Maps** (7) to create

standardized test scenarios. For evaluation, enter the **Model file** path (8) or browse (9), then click **Run Evaluation** (10). Progress appears in the bar (11) and detailed results show in the text area.

CLI and Script Reference

Although the primary tools are GUIs, you can also launch each module directly from the command line.

`UTILITY_LAUNCHER.py`

Desktop hub that opens other tools in separate processes.

```
python UTILITY_LAUNCHER.py
```

`simulator.py`

Main navigation simulator with pygame rendering.

```
python simulator.py
```

Interactive controls: - `W` / `S` or arrow keys: drive forward/backward - `A` / `D` or arrow keys: turn left/right - `Space`: toggle logging on/off - `R`: start new episode (Run) - `F`: fail current episode - `1`: switch to Keyboard input - `2`: switch to Gamepad input - `3`: switch to Model (autonomous) mode - `4`: Heading+Drive control mode - `5`: XY Strafe control mode - `6`: Hdg+Strafe control mode

`train_mlp.py`

Imitation learning trainer for MLPs.

```
python train_mlp.py
```

`train_ppo.py`

PPO reinforcement learning fine-tuning.

```
python train_ppo.py
```

`benchmark_gui.py`

Benchmark map generator and model evaluator.

```
python benchmark_gui.py
```

`headless_sim.py`

Non-GUI simulation environment (used by PPO trainer).

```
# Typically called internally by train_ppo.py
python headless_sim.py
```

Configuration

Simulator preferences (`simulator_prefs.json`)

The simulator saves user preferences to `simulator_prefs.json` in the project root. You can edit this file directly or use the **Preferences** dialog (button 10 in the simulator).

Setting	Description	Default
`history_len`	Number of timesteps in the input window (1–10)	1
`active_log_rate_hz`	Logging frequency when obstacles are near (1–30 Hz)	10.0
`min_turn_rate_dps`	Minimum turn rate in degrees/sec (0–39)	0.0
`inner_deadband_dps`	Deadband around zero turn rate (0–min_turn_rate)	0.0
`hysteresis_band_dps`	Hysteresis band to prevent jitter (0–min_turn_rate)	0.0
`obstacle_count`	Default number of obstacles per episode (1–50)	random 3–10
`vis_mode`	Visualization mode: `all`, `action_radius`, `detected`, `sparse_sensing`	`all`
`input_mode`	Default input: `keyboard`, `gamepad`, `model`	`keyboard`
`logging_enabled`	Whether logging is on by default	`false`
`robot_aligned_view`	Whether the view follows the robot heading	`true`
`control_mode`	Default control: `heading_drive`, `xy_strafe`, `heading_strafe`	`heading_drive`
`place_against_wall`	Place target against a wall	`false`

Model file format

Trained models are saved as JSON files with the following structure:

```
{
  "schema_version": 2,
  "mode": "heading_drive",
  "weights": [ [...], ...],      // List of weight matrices
  "biases": [ [...], ...],        // List of bias vectors
  "x_scale": { "mean": [...], "std": [...] },  // Input normalization
  "y_scale": { "mean": [...], "std": [...] },  // Output normalization
  "history_len": 1
}
```

Troubleshooting

Simulator fails to open

- Ensure `pygame` is installed (`pip install pygame`).
- Check that your display environment supports GUI applications (some headless servers require `xvfb`).

No gamepad detected

- Verify the gamepad is connected before launching the simulator.
- On Linux, you may need to install `joystick` package and run `jstest` to verify.

MLP training crashes with “shape mismatch”

- Make sure you are using logs generated with the same `history_len` as expected by the model.
- Re-train with matching settings or delete old logs.

PPO trainer shows “CUDA out of memory”

- Reduce **Parallel envs** (7) or **Minibatch size** (14).
- Switch **Device** (18) to `cpu` if you don’t have a compatible GPU.

Benchmark evaluation is slow

- Reduce the number of **# maps** (3) or obstacle counts.
- Evaluation runs sequentially; consider using fewer maps for quick checks.

Model fails to load (“Mode mismatch”)

- The model JSON was trained for a different control mode.
- Re-train a model for the desired mode or switch the simulator to the correct mode.

Generated by auto_labeler.